

RDI Testing

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Brief Introduction

Tester Data are diagnostic command used to:

- Check internal sw status and sensors
- Check actuators status
- Allow components learning (pedals, valves,...)

Internal SW Status, sensors reading etc, can be known in two ways:

- By **Local Identifier** – It is an ID defined during development phase. This is the most common approach used in ECUs in Stellantis programs and defines IDs for DIDs, IO Controls and Routine Controls;
- By **Memory Address** – It is an internal address where information to be read is stored.

In any case, many aspects such as conversion formulas, possibility to write by an external tool, data protection by security access, etc. are defined. All those definitions are developed in order to allow the vehicle to be serviced at the Plant (EOL) and in Aftersales and for development, fleet and validation activities.

What are DIDs? (1/2)

DID acronym stands for *Diagnostic ID* and identifies all engineering parameters generated by the sensors and processed by the electronic system as well as some parameters calculated and used for control and implementation that must be available on the diagnostic link.

These parameters will be made available on the diagnostic link, upon proper request, individually or in a group (with a specific command); in addition, they must be accessible in the key ON status, with the motor running and at any vehicle speed.

In case of errors on a certain component, the related parameter that is transmitted must be the one actually read by the sensor and not the recovery parameter used by the electronic system.

If one or more parameters are not present in an application (version, outfitting), upon request by the instrument of that specific parameter, the ECU shall respond with a negative response whose significance is “parameter not available” (**Nack 0x31**).

DIDs can be divided into two categories:

- **Readable DIDs** - In that case, DID can be also indicated with acronym RDI, which stands for Readable Diagnostic Identifier.
- **Writeable DIDs** - In that case, DID can be also indicated with acronym WDI, which stands for Writeable Diagnostic Identifier.
- **Both Readable and Writeable DIDs.**

What are DIDs? (2/2)

DIDs can be also divided in:

- **Mandatory DIDs**, as per diagnostic norms such as ISO 14229. Those DIDs must be implemented in proper ECUs and are generally indicated as **Identification parameters**. Generally, identification parameters are DIDs with F1xx identifier;
- **Engineering Parameters**, that are implemented for a certain ECU in a certain vehicle program. Those DIDs report informations about sensors, valves, pumps, pedals, ECU features and so on and their identifier is chosen from a specific range reserved to the customer and specified in CS.00102 (STLA world) like reported in the screen below:

DID type	Status	DID	Description
			Control service [2F hex] as well as the Read Data by DID service [22 hex])
Predefined FCA Specific Data Records	Reserved	6000 – 607F	Reserved for future Predefined FCA Specific Data Records
Predefined FCA Specific Data Records	Defined	6080	Aging Counter
Predefined FCA Specific Data Records	Defined	6081	Cycle To Switch Warning Indication OFF
Predefined FCA Specific Data Records	Reserved	6082-607F	Reserved for future Predefined FCA Specific Data Records assigned by FCA
Predefined FCA Specific Data Records	Defined	6090	Frequency Counter
Predefined FCA Specific Data Records	Reserved	6091 – 6FFF	Reserved for future Predefined FCA Specific Data Records assigned by FCA
Predefined FCA Specific Data Records	Defined	7000-7FFF	Open ECU specific DIDs

Conversion Formulas

DIDs, IOs and routines give informations about values of interest related to components linked to the ECU through CAN, LIN or hardwired and to learnings done.

Each diagnostic requirement has its own conversion formula to convert raw values in readable ones. There are different types of conversion formulas:

- **Linear conversion**, used to convert raw values in physical ones as following:

$$y(x) = m(x) + b$$

where $m(x)$ is the **conversion factor** and b is the **offset**. For each linear conversion, maximum and minimum values, both HEX and DEC, shall be specified, adding unit of measurement also;

- **Table conversion**, where each state assumed by the signal takes on a HEX value as in the example below:

Encoding Name	Mask	Min	Max
False	\$1	\$0	\$0
True	\$1	\$1	\$1

A mask specifies the maximum representable value based on the signal bit/byte size;

- **Identical conversion**, that is generally used to represent HEX or ASCII values as output.

Each conversion formula has its own ID, description and size.

02 – Service \$22/\$2E

Brief Introduction

Diagnostic Service \$22 allows reading the engineering parameter. In that case, DID can be also indicated with acronym RDI, which stands for Readable Diagnostic Identifier. As per UDS protocol, a DID can be read using following command:

\$22 \$XX XX

where \$XX XX is RDI identifier, generally implemented in 2 bytes size. Positive response given by the ECU is:

\$62 \$XX XX \$YY

where \$XX XX is always the RDI identifier and \$YY is the value you want to read.

Diagnostic Service \$2E allows writing values in the engineering parameter. In that case, DID can be also indicated with acronym WDI, which stands for Writeable Diagnostic Identifier. As per UDS protocol, a DID can be read using following command:

\$2E \$XX XX \$ZZ

where \$XX XX is WDI identifier, generally implemented in 2 bytes size and \$ZZ is the value you want to write. Positive response given by the ECU is:

\$6E \$XX XX

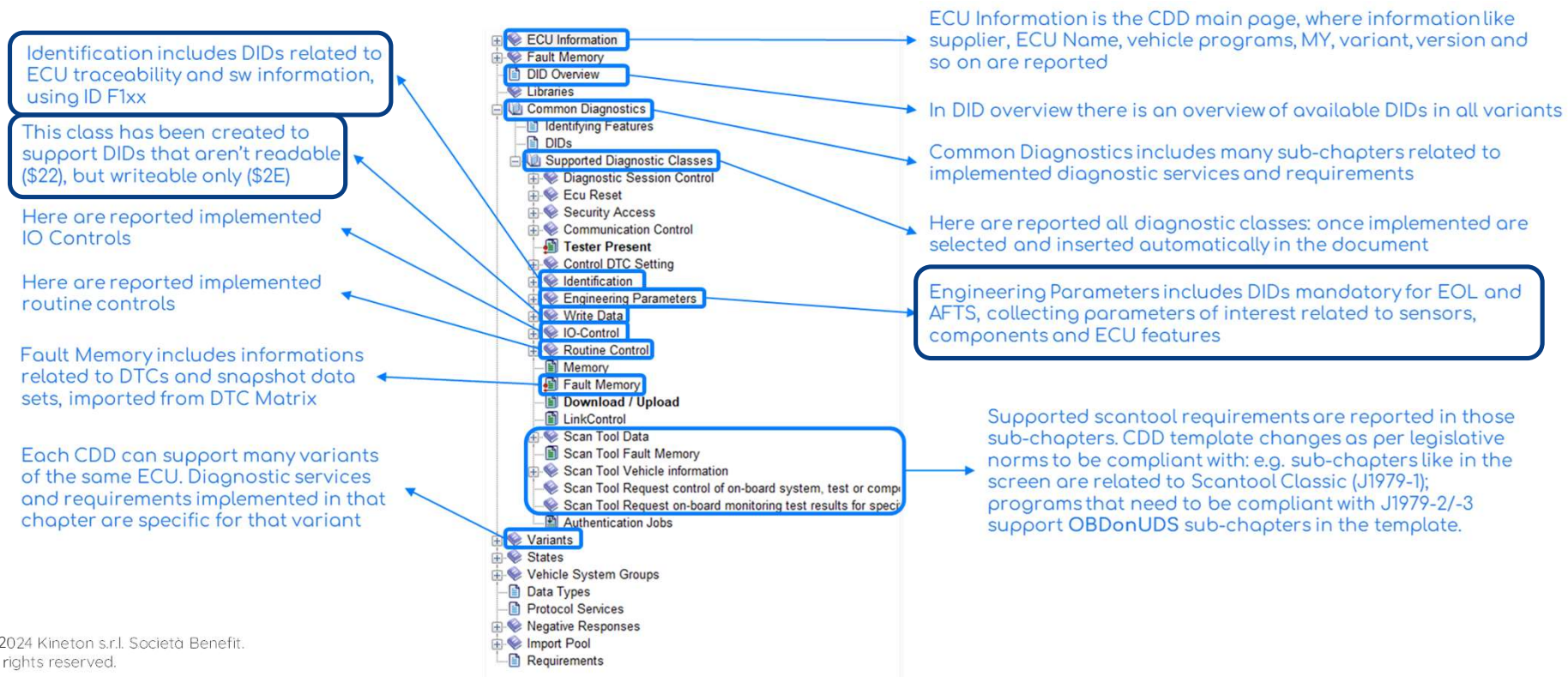
where \$XX XX is always the WDI identifier.



What is CANdelaStudio?

CDD (**CANdela Diagnostic Document**) is a diagnostic document developed with **CANdelaStudio**, an engineering tool from **Vector**. It contains all diagnostic information about the covered system.

There is a menu on the left with several chapters and sub-chapters:



03 – RDI in CDD

DID Overview

In *DID overview* there is an overview of available DIDs in all variants. It is such a list that reports DID identifier, DID description and if it is implemented in *Common Diagnostics* and/or in any variant:

DID Overview					
Here you see an overview of available DIDs in all variants:					
☒ Show DID Data ☒ Show Libraries Enter text to search...					
Identifier	Name	Common Diagnostics	EVCU_0004_00000020_021	Used	DID Data
0x0306	Accelerator Pedal Sensor 1	yes	yes	yes	Accelerator Pedal Sensor 1
0x0307	Accelerator Pedal Sensor 2	yes	yes	yes	Accelerator Pedal Sensor 2
0x0700	Main Processor Reset Source	yes	yes	yes	Main Processor Reset Source - Core 0, Main Processor Reset Source - Core 1
0x0702	PIM Internal Communication Failures	yes	yes	yes	HCP - MCPA SPI CRC Error Count, HCP - MCPA SPI ARC Error Count, MCPB - MCPA SPI CRC Error Count
0x0804	Immediate Stop Reasons	yes	yes	yes	Bitfield (Battery Contactor Status Not Closed, Immediate Stop Calibration Override, Power Mode Run/C
0x1000	Engine Speed	yes	yes	yes	Engine Speed
0x1002	Vehicle Speed	yes	yes	yes	Vehicle Speed
0x1004	Battery Voltage	yes	yes	yes	Battery Voltage
0x1006	Control Module Voltage	yes	yes	yes	Control Module Voltage
0x1008	ECU time stamp	yes	yes	yes	ECU Life Time
0x1009	ECU time stamp from ignition ON	yes	yes	yes	ECU Time Since Ignition ON
0x100A	Network Security Status	yes	yes	yes	Security Status, Authenticated Role
0x100B	Diagnostic Tool and Session Status	yes	yes	yes	New Bit Field (Tester Present Message, Default Session , Program Session, Extended Session, Other S
0x1206	ADC Control Module Voltage	yes	yes	yes	Battery Voltage
0x1921	Operational Mode Status	yes	yes	yes	Bitfield (Command Ignition Status #2 - Reserved (1), Command Ignition Status #2 - Reserved (2), Con



03 – RDI in CDD

DID Implementation

In the screen, DID name is reported and eventually also a description to include details related to diagnostic requirement

Here DID identifier is reported

Here services are enabled to specify if DID is readable or writeable or both. Information related to protocol service are available too

Diagnostic Instance (Engineering Parameters)

Name: Engine Speed

Description:

RecordDataIdentifier: 0x1000

Service:

☒ Read: ☐ Write:

Protocol Service:

[\(\\$22\) ReadDataByIdentifier](#)

[\(\\$2E\) WriteDataByIdentifier](#)

Request: 22 10 00 Pos. Resp.: 62 10 00 Neg. Resp.: 7F 22

2E 10 00 6E 10 00 7F 2E

DataRecord (zz)

Byte No.	Bit Pos.	Name	Data Type	Default	Constant	Desc
0		Engine Speed	From_0_to_1638375_RPM_2502_LC320			

Please click here to create a new element.

New Instance Delete Instance < Previous Next >

On the right there are informations related to the request made by the tester, positive and negative responses. If they're in grey, it means the service isn't enabled (e.g. service \$2E isn't enabled for DID \$1000 – Engine Speed, so it's readable only).

In DataRecord details related to the DID are reported like bytes/bit position, name and data type with conversion formula



03 – RDI in CDD

Properties of Data Type

Properties of Data Type is a screen that can be opened for all Tester Data to know more about conversion formulas.

DataRecord → click right on conversion formula name → Choose Properties of Data Type

LINEAR CONVERSION

The dialog box 'Properties of Data Type' for 'From_0_to_1638375_RPM_2502_LC320' is shown. The 'Linear' tab is selected. The 'Name' field contains 'From_0_to_1638375_RPM_2502_LC320'. The 'Length (Byte:Bit)' is set to 2 : 0. The 'Bitmask' is 0xFFFF. The 'ByteOrder' is set to 'HighLow (Motorola)'. The 'Encoding (raw)' is 'Unsigned' and 'Encoding (phys)' is 'IEEE Float (double)'. The 'Lower / Upper limit' section has 'Lower limit (raw)' at 0x00 and 'Upper limit (raw)' at 0xFFFF, with 'Lower limit (phys)' at 0 rpm and 'Upper limit (phys)' at 16383.75 rpm. The 'Conversion formula' section shows the formula: $phys = 0.25 / 1 \times data + 0$ with unit 'rpm'. The 'Display Format' is 'Floating Point' and 'Precision' is 2 decimal places. Buttons at the bottom include OK, Annulla, Applica, and ?.

TABLE CONVERSION

The dialog box 'Properties of Data Type' for 'RecordDataIdentifier' is shown. The 'Text Table' tab is selected. The 'Name' field contains 'RecordDataIdentifier'. The 'Bit length / field size' section has 'Length (Byte:Bit)' set to 2 : 0. The 'ByteOrder' is set to 'HighLow (Motorola)'. The 'Encoding' is 'Unsigned'. There are checkboxes for 'Discrete values' and 'Use inverse values'. A table with columns 'Start', 'End', 'Assigned Text', and 'Add. Informa...' is shown with one row: Start 0x00, End 0xFFFF. Buttons at the bottom include OK, Annulla, Applica, and ?. On the right side, there are buttons for 'Import...', 'New', and 'Delete'.

04 – RDI Testing

How to test RDIs?

RDI Testing is the most simple test to realize and it's available in default session \$10 01.

Below an example of a Test sequence:

TEST OK

\$22 \$XX \$XX

\$62 \$XX \$XX \$YY

TEST NOT OK

\$22 \$XX \$XX

\$7F \$22 \$11

Where:

- \$XX XX is RDI ID;
- \$YY is the HEX value read by the identifier.

Reason of Negative Response
(e.g. Service Not Supported)



Comparison with Internal Variable

Is only one reading enough to say the test is OK?

NO!

At least **3** readings are needed to say the test is OK and RDI is not blocked!

MAIN STEPS:

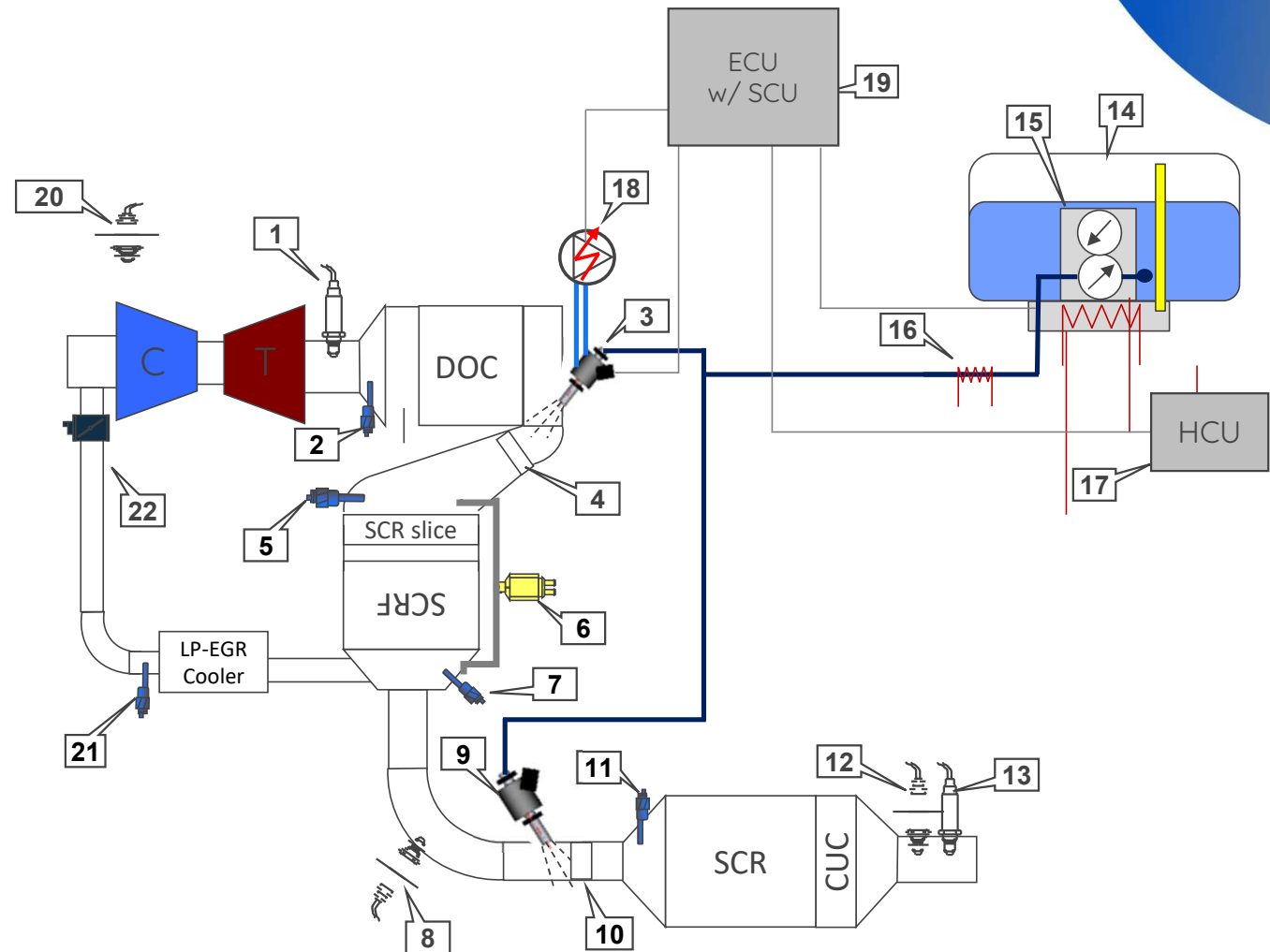
- Reading the RDI to know it's current value;
- Put the internal SW variable to the minimum value in the range and read the RDI to check if the modified value is readable;
- Put the internal SW variable to the maximum value in the range and read the RDI to check if the modified value is readable.

05 - Example

Test Case

- **ECU:** ECM
- **Sub-System:** SCR System
- **Issue:** All components in SCR system shall be tested in EOL to verify it is working properly
- **Customer request:** Implement DIDs to check all enable conditions to test SCR components are respected in EOL

Focus on → Urea Dosing Valve [9]



05 - Example

Test Execution in DIAnalyzer

Diagnostic Instance (Engineering Parameters)

Name: SCR System Dosing Valve 1 Preconditions

Description:

RecordDataId: 0x2409

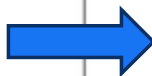
Service: ☒ Read: (\$22) ReadDataByIdentifier ☐ Write: (\$2E) WriteDataByIdentifier

Protocol Service:

DataRecord (zz)

Byte No.	Bit Pos.	Name	Data Type
0		SCR System Dosing Valve 1 Preconditions	HexDump (2)
0		SCR System Dosing Valve 1 Preconditions No Faults Present	True/False (1)
1		SCR System Dosing Valve 1 Preconditions Vehicle speed	Range/Out o
2		SCR System Dosing Valve 1 Preconditions Engine speed	Range/Out o
3		SCR System Dosing Valve 1 Preconditions Catalyst temperature	Range/Out o
4		SCR System Dosing Valve 1 Preconditions Tank temperature	Range/Out o
5		SCR System Dosing Valve 1 Preconditions Tank level	Range/Out o
6		SCR System Dosing Valve 1 Preconditions Battery voltage	Range/Out o
7		SCR System Dosing Valve 1 Preconditions First Filling Has Not Been Performed Yet	True/False (1)
8		SCR System Dosing Valve 1 Preconditions Hydraulic System in Unpressurized State	True/False (1)
9		Bit 9 not used	Reserved (1)
10		SCR System Dosing Valve 1 Preconditions Key run	True/False (1)
11		Bit 11 not used	Reserved (1)
12		Bit 12 not used	Reserved (1)
13		Bit 13 not used	Reserved (1)
14		Bit 14 not used	Reserved (1)
15		Bit 15 not used	Reserved (1)

Please click here to create a new element.



Stop Default Clear Errors

Messages: ReadDataByIdentifier

Signals: Value:

Addr: Descr: 3 byte: 22 00 00 ...

none 22 24 09 Send TX File

Tx every [ms]: 100

Normal Decode short

```
Bit 13 not used = Plausible value
Bit 14 not used = Plausible value
Bit 15 not used = Plausible value
)
26756.03340 f1->10 Tx[3] 22 24 09 : ReadDataByIdentifier
26756.15233 Error -2011: Rx Timeout
26771.01763 f1->10 Tx[3] 22 24 09 : ReadDataByIdentifier
26771.04672 10->f1 Rx[5] 62 24 09 01 20 : RDI_2409 SCR System Dosing Valve 1 Preconditions Read :
SCR System Dosing Valve 1 Preconditions = {
  SCR System Dosing Valve 1 Preconditions No Faults Present = False
  SCR System Dosing Valve 1 Preconditions Vehicle speed = in range
  SCR System Dosing Valve 1 Preconditions Engine speed = in range
  SCR System Dosing Valve 1 Preconditions Catalyst temperature = in range
  SCR System Dosing Valve 1 Preconditions Tank temperature = in range
  SCR System Dosing Valve 1 Preconditions Tank level = out of range
  SCR System Dosing Valve 1 Preconditions Battery voltage = in range
  SCR System Dosing Valve 1 Preconditions First Filling Has Not Been Performed Yet = False
  SCR System Dosing Valve 1 Preconditions Hydraulic System in Unpressurized State = True
  Bit 9 not used = Plausible value
  SCR System Dosing Valve 1 Preconditions Key run = False
  Bit 11 not used = Plausible value
  Bit 12 not used = Plausible value
  Bit 13 not used = Plausible value
  Bit 14 not used = Plausible value
  Bit 15 not used = Plausible value
}
```





Thank you!

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